

LINA -Lab03– Trigonometric Formulas

Work individually and make sure that throughout the course of this lab, you re-save sufficiently your solutions file on your own machine, finally delivered **.PDF**

LINA_2627_IGP99E_Van_Der_Zever_Johnny_Lab**03**.PDF (< 5 MB),

Reduce via https://www.ilovepdf.com/compress_pdf or Adobe Acrobat (Online)

→ which is to submit via week-folder Leho/ LINA / Week**03**/Lab**03** Practice

Also, make sure to repeat Class group + Full Name +Lab**03** number inside it.

1. Exercises

1.1. Basic exercises

Please crack these basic exercises purely pen and paper, consulting theory either from your notes and/or your book (chapter 3).

1.1.1. Angles

Opposite and complementary angles

A drawing will help you to solve these questions:

- a) What is the result of $\sin(\alpha) + \sin(\beta)$ when α and β are opposite angles?
- b) What is the result of $\sin^2(\alpha) + \sin^2(\beta)$ when α and β are complementary angles? A drawing helps to find the solution.
- c) What is the result of $\cos^2(\alpha) + \cos^2(\beta)$ when α and β are complementary angles?

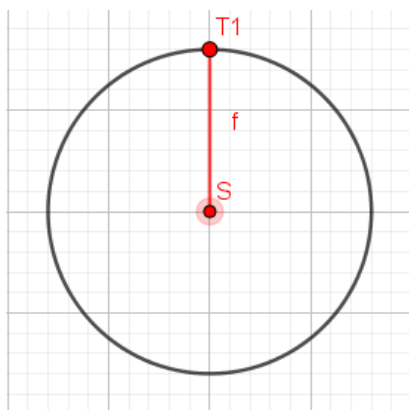
1.1.2. Rotation

Position a plane point $P(x,y)$ at a fixed distance r to the origin O . Now, rotate this point around the origin O over an angle β along a circular arc (i.e.: keeping its distance r to the origin).

Determine the rotated point $P'(x',y')$ **coordinates** in terms of the initial coordinates x and y .

Hint: consider the initial point $P(x,y)$ subtending an angle α to the horizontal x -axis.

1.1.3. Clock

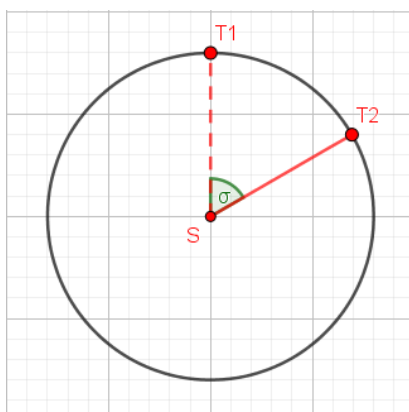


A game shows a chronometer that measures the time an attack takes.

The arm starts upwardly as indicated by [ST1].

And it makes a whole tour in 1 minute.

It has a diameter of 16 units.



T2 indicates the position of the arm after 10 seconds. Answer the questions mentioned in the table below.

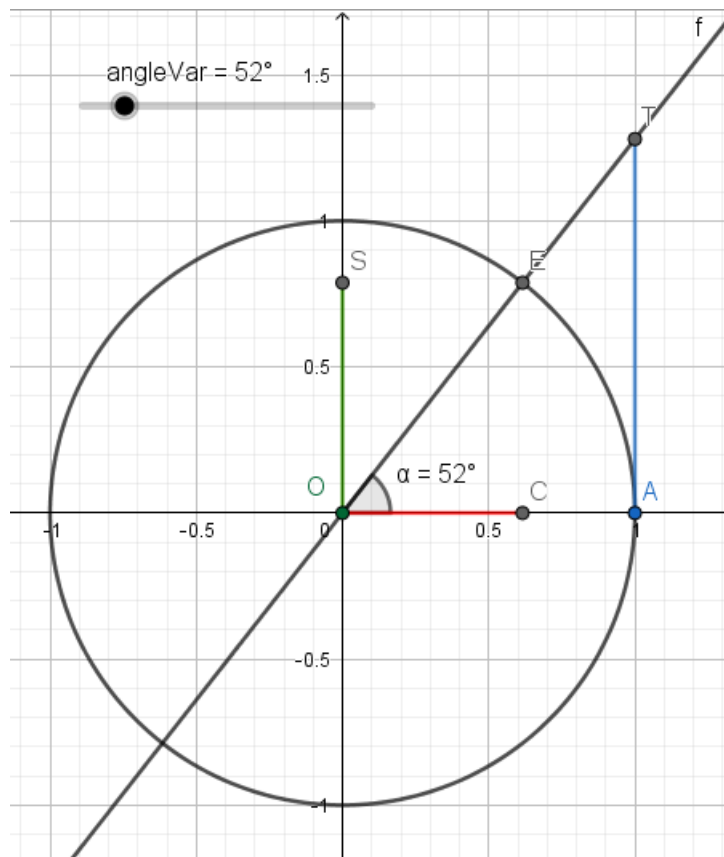
Fill the first column 'Formula' of this table purely pen and paper.

	Formula	Result
a) Size in degrees of angle σ		
b) Horizontal distance between S and T2		
c) Vertical distance between S and T2		

Compute from the first column, the second 'Result' in **GeoGebra Classic 6**.

Finally, also verify your complete table by a drawing in GeoGebra Classic 6.

1.1.4. Exploring trigonometric ratios in the unit circle



Open the given file

LINA-Lab03_TrigonometricFormulas(UnitCircle).ggb

The graphics view shows a unit circle and a line that forms an angle α with the positive x. The angle has the same value as the slider value, so it changes when you move the slider. The Algebra view shows the values of the cosine, sine and tangents of this angle. The points C, S and T also correspond with these values.

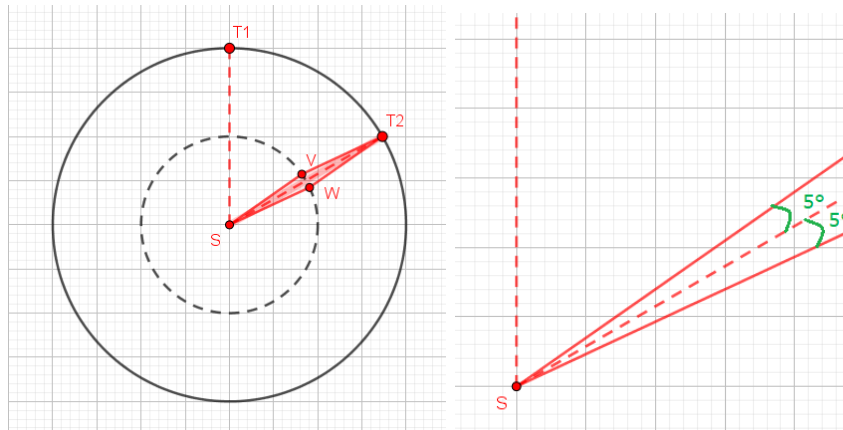
- For which angle values is the tangent undetermined?
- Give the angle interval(s) that results in negative cosine values?
- Give the angle interval(s) that results in negative sine values?
- Give the angle interval(s) that results in negative tangent values?
- What is the maximum and minimum value of the cosine and sine?
- Which angles have the same cosine and sine value?

1.2. Contextual practice

Now solving following more advanced problems should become feasible.

1.2.1. Advanced clock

The project leader wants you to draw a more advanced arm.



The vertices V and W of the arm are on a circle with diameter half the value (8 units) of the outer circle (16 units) and the line segments [SV] and [SW] form an angle of 5° to ST2.

What is the horizontal and vertical distance between S and V and S and W after 10 seconds?

	Formula	Result
a) Horizontal distance between S and V		
b) Vertical distance between S and V		
c) Horizontal distance between S and W		
d) Vertical distance between S and W		

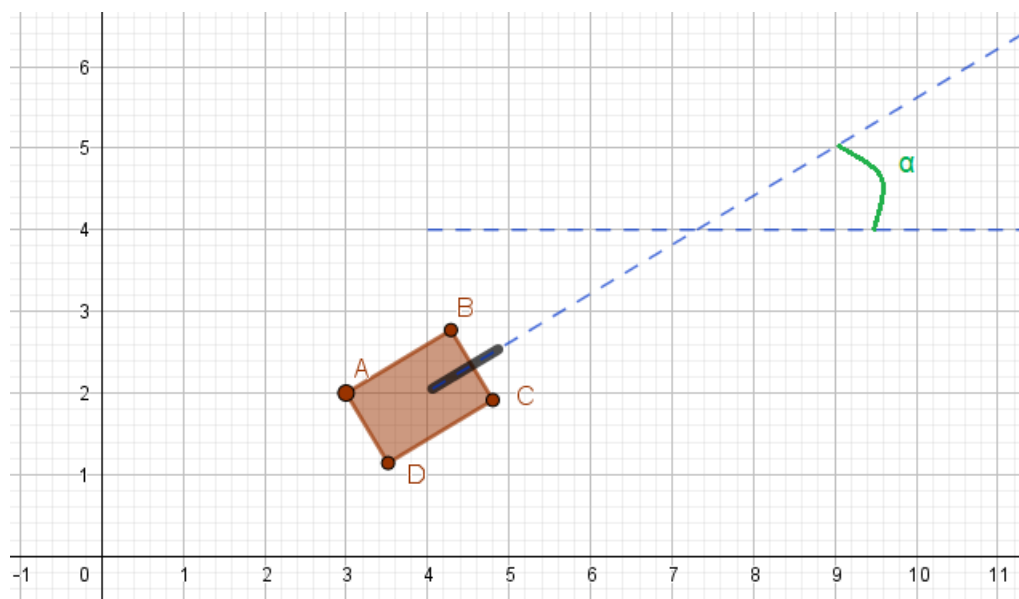
Now construct in GeoGebra Classic 6 a working animation of this clock's arm. Make you sure mimic the shape of it and insert a slider to turn this arm around.

1.2.2. Tank wars

We are developing a tank wars game.

The image below shows the top view of one of the tanks. It has a rectangular form of which the diagonal $[AC]$ lies *nearly* horizontal. The coordinates of vertex A are $(3,2)$ and segments $[AD]$ and $[BC]$ measure 1 unit each. $[AB]$ and $[DC]$ both measure 1.5 units. The tank its gun is aligned with the tank and pointing in the direction determined by the dashed line with angle α . The slope ($=\tan(\alpha)$) of this direction equals $+0.6$.

Calculate the coordinates of the vertices B , C and D of the tank.

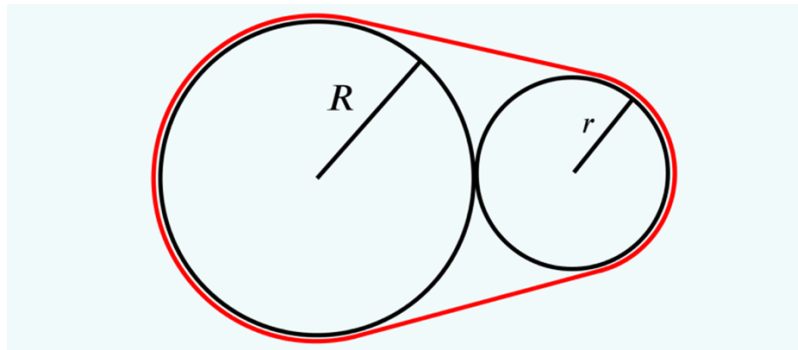


	Formula	Result
a) x of B		
b) y of B		
c) x of D		
d) y of D		
e) x of C		
f) y of C		

Now construct in GeoGebra Classic 6 a working animation of this tank. Make you sure mimic the shape of it and insert sliders to make it ride.

1.2.3. Belt

The red belt keeps together two wheels of radii R and r respectively, looping all the way around them both. We logically assume $R > r > 0$.



Determine the entire **length** of this looped belt.

Hint 1. Distinguish the four parts of this belt: two external tangent stretches, the left circular arc and the right circular arc. Calculating each of these partial lengths relies on properties of the tangent to a circle, right triangles and (inverse) trigonometry.

Hint 2. You will need to solve a certain right triangle and will need to apply the definition of an angle in radian.

2. References

2.1. Trigonometry

ISBN 9789401474955 (Animation Maths NE/2021), **pars 3.4 – 3.8**

2.2. Geogebra tutorials

2.2.1. The basics

<https://www.youtube.com/watch?v=1cBXWi66-tY>

2.2.2. Sine and Cosine in a Unit Circle

<https://www.youtube.com/watch?v=H-MTLHRkcn8>